

# ARRAYS

## Q2) 2D Array Problem :

### Question:

| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Session | Arrays | Question Information | Level 1 | Challenge 22 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|----------------------|---------|--------------|
| <p>Problem Description: Ram has provide inputs two numbers 'p' and 'q' to Sakthi. He wants to creates a matrix of size p x q (p rows and q columns) in which every elements is either Y or O. The Ys and Os must be filled alternatively, the matrix should have outermost rectangle of Ys, then a rectangle of Os, then a rectangle of Ys, and so on..</p> <p>Constraints:</p> <p>1 &lt;= p, q &lt;= 1000</p> <p>Input Format:</p> <p>Input lines must be how many rows and columns in that matrix, also values must be separate space.</p> <p>Output Format:</p> <p>Print the output in a separate lines.</p> |         |        |                      |         |              |

## PROGRAM:

```
Arrays > C Q2.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      int p, q;
6
7      printf("Input:");
8      scanf("%d %d", &p, &q);
9      printf("\n");
10
11     printf("Output");
12     printf("\n");
13
14     for (int i = 0; i < p; i++)
15     {
16         for (int j = 0; j < q; j++)
17         {
18             int top = i;
19             int left = j;
20             int bottom = p - 1 - i;
21             int right = q - 1 - j;
22
23             int layer = top;
24
25             if (left < layer)
26                 layer = left;
27
28             if (bottom < layer)
29                 layer = bottom;
30
31             if (right < layer)
32                 layer = right;
33
34             if (layer % 2 == 0)
35                 printf("Y ");
36             else
37                 printf("0 ");
38         }
39
40         printf("\n");
41     }
42
43     return 0;
44 }
```

## OUTPUT:

### TEST CASE 1:

Input: 6 7

Output

```
Y Y Y Y Y Y Y
Y 0 0 0 0 0 Y
Y 0 Y Y Y 0 Y
Y 0 Y Y Y 0 Y
Y 0 0 0 0 0 Y
Y Y Y Y Y Y Y
```

### TEST CASE 2:

Input: 5 8

Output

```
Y Y Y Y Y Y Y Y
Y 0 0 0 0 0 0 Y
Y 0 Y Y Y Y 0 Y
Y 0 0 0 0 0 0 Y
Y Y Y Y Y Y Y Y
```